

Static Properties aur Methods

Static method kya hai?

Wo method jo **poori class** ka hota hai, kisi ek object ka nahi. `static` keyword lagate hain.

```
javascript

class User {
  static staticMethod() {
    console.log(this === User); // true
  }
}

User.staticMethod();
```

Ye `User.staticMethod = function...` likhne jaisa hi hai. Andar `this` = class khud (dot se pehle wala rule).

Kab use hota hai?

1. Compare/helper functions:

```
javascript

class Article {
  constructor(title, date) { this.title = title; this.date = date; }
  static compare(a, b) { return a.date - b.date; }
}

articles.sort(Article.compare);
```

2. Factory methods (alag tareeke se object banane ke liye):

```
javascript

static createToday() {
  return new this("Today's digest", new Date());
}
```

3. Database jaisi classes me search/save/remove.

Dhyan: static methods sirf class par chalte hain, object par nahi.

```
javascript

article.createToday(); // Error
```

Static properties

javascript

```
class Article {  
  static publisher = "Ilya Kantor";  
}  
Article.publisher;
```

Class-level data ke liye (kisi instance se nahi jura).

Static ki inheritance

Static properties/methods **inherit** hote hain.

javascript

```
class Animal {  
  static planet = "Earth";  
  static compare(a, b) { return a.speed - b.speed; }  
}  
class Rabbit extends Animal {}  
  
Rabbit.planet;    // Earth  
Rabbit.compare;   // Animal.compare
```

Kaise? `extends` do prototype links banata hai:

1. `Rabbit.__proto__ === Animal` (statics ke liye)
2. `Rabbit.prototype.__proto__ === Animal.prototype` (normal methods ke liye)

Task: `class Rabbit extends Object`

`class Rabbit`

`class Rabbit extends Object`

kuch nahi

constructor me `super()` zaroori

`Rabbit.__proto__ === Function.prototype`

`Rabbit.__proto__ === Object`

Matlab `extends Object` karne par `Rabbit.getOwnPropertyNames(...)` jaise `Object` ke static methods bhi mil jaate hain.

Yaad rakho

- `static` = class ka apna kaam/data.
- `Object` par nahi, class par call karo.

- Child class ko statics parent se mil jaate hain.